INFOBuddy AI Chatbot

Project Overview

INFOBuddy is an **AI-powered educational chatbot** designed to assist users by providing information and answering queries using Natural Language Processing (NLP). The chatbot can understand user questions and deliver relevant responses across different knowledge domains.

Problem Statement

Users often spend significant time searching for educational resources online. An intelligent chatbot can provide quick answers and personalized assistance.

Objectives

- Build an AI chatbot capable of understanding natural language.
- Provide accurate and relevant responses.
- Assist users with educational queries.

Technologies Used

Programming Language – Python
Natural Language Processing - NLTK, spaCy, Transformers
AI Framework - LangChain
Backend Development - Flask, FastAPI
Frontend Development - HTML, CSS, JavaScript
Data Storage - Firebase, JSON
Tools & Environment - GitHub, Google Colab

Methodology

1. User inputs a question.
2. NLP processes the query.
3. The chatbot identifies intent and context.
4. A response is generated using AI models.

Key Features

- Natural language query understanding
- Context-based responses
- Multi-domain knowledge assistance
- Interactive conversational interface

Results

INFOBuddy can efficiently respond to educational queries and assist users in accessing relevant information quickly.